



MAKERERE

UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SYSTEM

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

WEEK 4: TOOL CATALOGUE & API SCHEMAS

PROJECT: University Student-Support Case Agent

STUDENT DETAILS:

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Executive Summary

Purpose. This document specifies the tools, input/output schemas, failure behaviour, escalation rules, Gemini function-calling declarations, and architectural alignment for the University Student-Support Case Agent.

Safety boundary. The model does not directly modify databases or make authoritative academic decisions. Tool calls are constrained by deterministic schemas, parameter validation, and an authorization policy layer.

Tools covered. Two primary tools are defined: **get_case_status** for read-only case lookup, and **create_support_ticket** for structured ticket creation with human approval for high-impact cases.

Document Structure

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1. Overview and Purpose

In accordance with the BSE4104 Capstone Project Brief (Week 4: Tools and Function Calling), the Student-Support Case Agent moves beyond text generation to execute explicit, safe, and inspectable software tools.

The design maintains the following boundaries:

- The model never makes direct database modifications or authoritative academic decisions.
- All tool executions pass through deterministic schemas, parameter validation gates, and an authorization policy layer.
- Two primary tools are defined: get_case_status and create_support_ticket.
- get_case_status is a deterministic, read-only lookup tool querying pre-existing case records.
- create_support_ticket creates a structured, pending support ticket in the mock registry and is subject to human-in-the-loop approval for high-impact issues.

2. Tool 1: get_case_status

2.1 Description and Purpose

Retrieves real-time status and metadata for an existing student support case using a validated Case ID. This prevents hallucinated case progress updates and ensures that status information presented to students is factual and sourced directly from the departmental case database.

2.2 Operational Specification.

Property	Specification
Side Effects	None (read-only query).
Expected Latency	< 50ms (local sandbox store).
Authorization Scope	Open to authenticated/active student sessions querying their own case.

2.3 Input Schema (JSON Schema / OpenAPI)

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "GetCaseStatusInput",
  "type": "object",
  "required": ["case_id"],
  "properties": {
    "case_id": {
      "type": "string",
      "description": "The unique support case/ticket identifier in the format CAS-YYYY-XXX or TCK-YYYY-XXXX (e.g. CAS-2026-001, TCK-2026-0001).",
      "pattern": "^(CAS-\\d{4}-\\d{3}|TCK-\\d{4}-\\d{4})$",
      "examples": ["CAS-2026-001", "TCK-2026-0001"]
    }
  },
  "additionalProperties": false
}
```

2.4 Output Schema (JSON Schema / OpenAPI)

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "GetCaseStatusOutput",
  "type": "object",
  "required": ["case_id", "status", "category", "assigned_officer", "last_updated", "summary"],
  "properties": {
    "case_id": { "type": "string" },
    "student_id": { "type": "string", "pattern": "^\\d{10}$" },
    "category": {
      "type": "string",
      "enum": ["REGISTRATION", "EXAMINATION", "TIMETABLE", "GENERAL_QUERY"]
    },
    "status": {
      "type": "string",
      "enum": ["OPEN", "IN_PROGRESS", "RESOLVED", "PENDING_STAFF_APPROVAL", "CLOSED"]
    },
    "assigned_officer": { "type": "string" },
    "last_updated": { "type": "string", "format": "date-time" },
    "summary": { "type": "string" },
    "official_notes": { "type": "string" }
  }
}
```

2.5 Failure Behaviour and Error Codes

Error Code	HTTP Status	Description	System Response / Agent Handling
CASE_NOT_FOUND	404	Case ID does not exist in the registry.	Inform the student the ID was not found; prompt them to double-check the ID or create a new ticket.
INVALID_ID_FORMAT	400	Provided ID does not match the expected case-ID format.	Prompt the student to provide the ID in the standard format CAS-YYYY-XXX.
DATABASE_TIMEOUT	503	Database lookup timed out or service is unavailable.	Inform the student that the case lookup service is temporarily unreachable; do not fabricate a status.

3. Tool 2: create_support_ticket

3.1 Description and Purpose

Creates a new support ticket in the mock department ticketing database when procedural RAG self-service cannot resolve the student's problem. The model extracts messy, free-text student statements into normalized, structured parameters.

3.2 Operational Specification.

Property	Specification
Side Effects	State creation: appends a new ticket record in the mock database.
Expected Latency	< 100ms.
Authorization Scope	Student user; high-impact or sensitive categories automatically route to human approval.

3.3 Input Schema (JSON Schema / OpenAPI)

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "CreateSupportTicketInput",
  "type": "object",
  "required": ["student_id", "category", "summary", "details", "urgency"],
  "properties": {
    "student_id": {
      "type": "string",
      "description": "The student's 10-digit Makerere student number (e.g. 2300708510).",
      "pattern": "^\\d{10}$"
    },
    "category": {
      "type": "string",
      "description": "The normalized problem classification.",
      "enum": ["REGISTRATION", "EXAMINATION", "TIMETABLE", "GENERAL_QUERY"]
    },
    "summary": {
      "type": "string",
      "description": "Concise single-sentence summary of the student's issue.",
      "minLength": 5,
      "maxLength": 100
    }
  }
}
```

```

    },
    "details": {
      "type": "string",
      "description": "Relevant specifics, course codes, or problem context extracted from the student's description.",
      "minLength": 10,
      "maxLength": 500
    },
    "urgency": {
      "type": "string",
      "description": "Urgency classification.",
      "enum": ["LOW", "MEDIUM", "HIGH"]
    }
  },
  "additionalProperties": false
}

```

3.4 Output Schema (JSON Schema / OpenAPI)

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "CreateSupportTicketOutput",
  "type": "object",
  "required": ["ticket_id", "status", "created_at", "routing_queue",
    "requires_human_approval"],
  "properties": {
    "ticket_id": { "type": "string", "pattern": "^TCK-\\d{4}-\\d{4}$" },
    "status": {
      "type": "string",
      "enum": ["OPEN", "PENDING_STAFF_APPROVAL"]
    },
    "created_at": { "type": "string", "format": "date-time" },
    "routing_queue": { "type": "string" },
    "requires_human_approval": { "type": "boolean" },
    "acknowledgment_message": { "type": "string" }
  }
}

```

3.5 Human-in-the-Loop Escalation Rules

The tool applies the following deterministic escalation rule:

Condition	Result
category == "EXAMINATION" OR urgency == "HIGH"	status = "PENDING_STAFF_APPROVAL"
Same condition	requires_human_approval = true
Same condition	routing_queue = "FACULTY_REGISTRAR_TRIAGE"
Student-facing acknowledgement	Your ticket has been logged and flagged for priority review by department staff before action is taken.

This rule ensures that examination-related or high-urgency cases do not proceed as unreviewed automated actions.

3.6 Failure Behaviour and Error Codes

Error Code	HTTP Status	Description	System Response / Agent Handling
VALIDATION_FAILED	422	Missing or malformed field, such as an invalid student number.	Ask the student for the missing/corrected information before invoking the tool.
DISALLOWED_TOPIC	403	Attempt to log a ticket demanding grade altering, fee cancellation, or disciplinary dismissal.	Tool rejects creation and triggers a boundary redirect message pointing to the official human board.
DUPLICATE_TICKET	409	Identical unresolved ticket already logged by this student.	Alert the student of the existing active ticket and provide the matching ticket ID.

4. Gemini Function Calling Declarations

The tool definitions above map to Google GenAI / Gemini Python tool declarations used in `src/model_client.py`.

4.1 get_case_status Declaration

```
GET_CASE_STATUS_DECLARATION = {
    "name": "get_case_status",
    "description": "Look up an existing student support case by its Case ID to get its current status, category, and official notes.",
    "parameters": {
        "type": "OBJECT",
        "properties": {
            "case_id": {
                "type": "STRING",
                "description": "The case ID formatted as CAS-YYYY-XXX (e.g. CAS-2026-001).",
            }
        },
        "required": ["case_id"]
    }
}
```

4.2 create_support_ticket Declaration

```
CREATE_SUPPORT_TICKET_DECLARATION = {
    "name": "create_support_ticket",
    "description": "Create a new support ticket for a student issue when procedural guidance cannot resolve it.",
    "parameters": {
        "type": "OBJECT",
        "properties": {
            "student_id": {
                "type": "STRING",
                "description": "The 10-digit student ID number (e.g. 2300708510).",
            },
            "category": {
                "type": "STRING",
            }
        }
    }
```

```

    "enum": ["REGISTRATION", "EXAMINATION", "TIMETABLE", "GENERAL_QUERY"],
    "description": "Problem classification category."
  },
  "summary": {
    "type": "STRING",
    "description": "Brief one-sentence summary of the problem."
  },
  "details": {
    "type": "STRING",
    "description": "Detailed explanation of the issue."
  },
  "urgency": {
    "type": "STRING",
    "enum": ["LOW", "MEDIUM", "HIGH"],
    "description": "Urgency level of the support request."
  }
},
"required": ["student_id", "category", "summary", "details", "urgency"]
}
}

```

5. Architectural Alignment Matrix

User Story	Associated Tool	Expected Deterministic Behaviour	Model Responsibility
US-2 — Check case status	get_case_status	Returns exact database status field; never invents status if lookup fails.	Parses case ID from user message and formats returned status into the student reply.
US-4 — Raise ticket	create_support_ticket	Creates record with unique ticket_id, sets initial timestamp and queue.	Extracts category, summary, details, and urgency from student description.
US-6 — Staff receives summarized ticket	create_support_ticket	Saves structured parameters directly into support staff review queue.	Normalizes colloquial slang or verbose student complaints into concise technical summaries.
US-10 — Refuse to invent status	get_case_status	Throws explicit error code upon timeout or missing ID.	Plainly states failure to the student rather than guessing a status.

6. Implementation and Review Checklist

- Implement both tools with the exact names and required parameters specified in this document.
- Validate inputs before any tool invocation.
- Enforce authorization so students can query their own case only.
- Keep get_case_status strictly read-only.
- Apply human-in-the-loop escalation for examination-related or HIGH-urgency tickets.
- Return explicit error codes for validation, missing records, service failure, disallowed topics, and duplicate tickets.
- Ensure the model never invents case status when the lookup fails.
- Keep Gemini function-calling declarations synchronized with the tool schemas.
- Verify alignment against US-2, US-4, US-6, and US-10.